

## Summary of Findings and Claims

### 1. STITCH: Structured Intent Tracking in Contextual History

Findings: STITCH is a memory system that enhances large language models' performance in long-horizon, goal-oriented interactions. It addresses context-mismatched evidence retrieval by indexing each step of a trajectory with structured retrieval cues and contextual intent.

Claims:

Reduces retrieval noise and improves long-horizon reasoning.

Outperforms the best baseline by 35.6%, especially with longer trajectories.

Introduces CAME-Bench for evaluating context-aware retrieval.

### 2. Impact of Generative AI on Architectural Design Tasks

Findings: The study shows that generative AI (GenAI) does not provide an overall performance advantage but significantly improves design performance for novice designers. Cognitive load remains similar across conditions, with some prompting strategies reducing cognitive load.

Claims:

GenAI's effectiveness is influenced by users' prior expertise and interaction strategies.

General creative self-efficacy decreases for students using GenAI.

### 3. Conversational Exam Format for Assessment

Findings: The proposed conversational exam format allows students to code live while explaining their reasoning, aiming to restore assessment validity in the context of generative AI use.

Claims:

This method is scalable and emphasizes authentic practice.

Real-time performance cannot be faked, enhancing assessment integrity.

### 4. Scaling Laws in Neural Networks

Findings: The paper investigates scaling laws in transformers trained on random walks, revealing consistent scaling exponents even without power law structures in data correlations.

Claims:

Key results in language modeling can be replicated using simpler models.

Proposes a new approach for optimal compute curves and critiques previous fitting methods.

## 5. PACEvolve Framework for Evolutionary Search

Findings: PACEvolve addresses challenges faced by large language models in evolutionary search, including context pollution and mode collapse, enabling consistent self-improvement.

Claims:

Achieves state-of-the-art performance on various benchmarks.

Incorporates hierarchical context management and momentum-based backtracking.

## References

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